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CHAPTER 1 RESULTS

This chapter presents the analytical outputs of the in silico phenotyping pipeline described in Chapter **??**. Section 1.1 describes the per-modality cohort composition following data retrieval and label matching. Section 1.2 documents the feature-retention and quality-control steps applied to each platform. Section 1.3 establishes the clinical, immunological, and genomic properties of the subtypes, establishing biological context of the subtypes. Section 1.4 quantifies the geometric separability of the subtypes within each omics feature space using Principal Component Analysis (PCA), Silhouette scoring, and Permutational Multivariate Analysis of Variance (PERMANOVA).

1.1 Cohort Characteristics

TCGA-LUSC datasets for five modalities were retrieved and matched to molecular subtype labels (described in Section **??**). Gene expression and Copy Number Variation (CNV) retained the full labelled 178-sample cohort following patient identifier harmonisation, whereas Deoxyribonucleic acid (DNA) methylation, micro RNA (miRNA) expression and Reverse Phase Protein Assay (RPPA) had smaller assay overlap resulting in cohorts of 76, 107, and 112 labelled samples respectively (Table 1.1). Differences in cohort size across modalities reflects differences in assay coverage across The Cancer Genome Atlas (TCGA) patients.

Across all modalities, classical subtype was the most frequently represented class (n = 29-65 depending on modality) and primitive the least frequent (n = 11-27), motivating the use of macro-averaged classification metrics in subsequent analyses to ensure equal weighting of each subtype despite varying sample sizes.

Table 1.1: Summary of omics modalities used in this study. Sample counts reflect the subset of samples with both available assay data and matching molecular subtype labels. Variation in total sample size across modalities reflects differential assay coverage.

Modality	**Configuration**	**Classical**	**Basal**	**Secretory**	**Primitive**	**Total**
Gene expression	HiSeqV2, RNASeqV2 RSEM-normalised	65	43	43	27	178
DNA methylation	HumanMethylation450k beta values	29	14	22	11	76
miRNA expression	miRNA sequencing, log2(RPM + 1)	44	29	20	14	107
RPPA	Normalised protein abundance	38	28	28	18	112
CNV	GISTIC2 all-thresholded calls	65	43	43	27	178

1.2 Preprocessing Summary

Modality-specific preprocessing was performed to respect the native data format and biological meaning of each assay (described in Section ??). The feature-retention summary is provided in Table 1.2, and per-modality Median Absolute Deviation (MAD) distributions for retained and discarded features are shown in Figure 1.1.

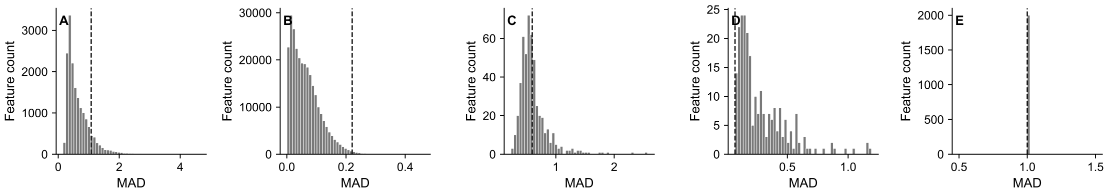


Figure 1.1: Feature variability distribution across processed omics modalities. Panels show MAD distributions for (A) gene expression, (B) DNA methylation, (C) miRNA expression, (D) RPPA, and (E) CNV. Dashed vertical lines denote MAD value of the last retained-feature threshold used for the processed matrix; for RPPA, all 238 antibodies were retained, the line denotes the minimum observed MAD among retained antibodies.

Table 1.2: Preprocessing and feature-retention summary for each omics modality. Raw feature counts correspond to the input assay feature spaces prior to any filtering. MAD threshold denotes the MAD score of the last retained feature in the processed matrix; for RPPA, where all 238 antibodies were retained, this value represents the minimum observed MAD among all retained antibodies.

Modality	**Raw features**	**Retained features**	**MAD threshold**
Gene expression	60,660	2,000	1.0808
DNA methylation	485,577	2,000	0.2208
miRNA expression	1,881	200	0.5954
RPPA	238	238	0.0692
CNV	24,776	2,000	1.0000

For gene expression, RNA-Seq by Expectation Maximization (RSEM)-normalised read counts provided in $\log_2(x + 1)$-transformed form by University of California, Santa Cruz (UCSC) Xena required no further normalisation. A low-expression filter (median $\log_2$ expression < 1.0 across all samples) removed 3,440 unexpressed or near-unexpressed transcripts from 20,530 candidates, retaining 17,090 expressed genes, from which the 2,000 most variably expressed were selected by MAD (MAD threshold: 1.08). For DNA methylation, beta values on the interval $[0, 1]$ were used directly without transformation. Of 485,577 Illumina Infinium HumanMethylation450 probes, 90,052 were removed for exceeding a per-probe missingness threshold of 20%, and a further 119,054 were removed as near-invariant (standard deviation < 0.05); the top 2,000 most variable probes were subsequently retained by MAD (threshold: 0.22). The lower MAD threshold for methylation relative to gene expression reflects the bounded $[0, 1]$ scale of beta values, which compresses absolute variability. Missing values in the methylation matrix following high-missingness probe removal were sparse (probe-level failures) and were imputed using per-probe medians (Figure 1.2).

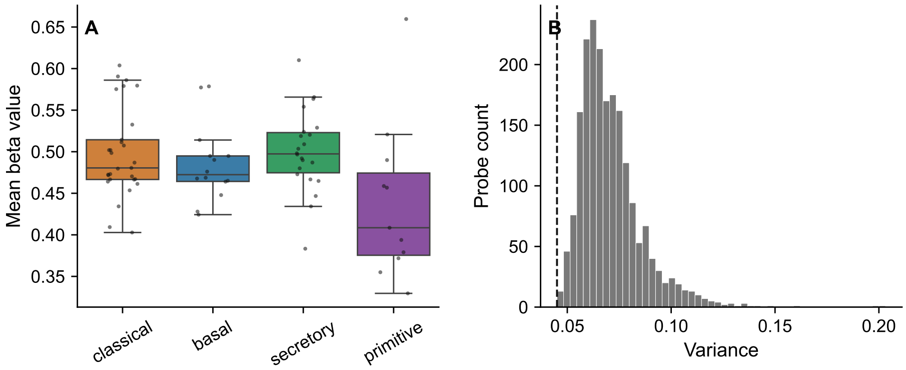


Figure 1.2: DNA methylation overview for labelled samples. (A) Mean beta value per sample stratified by subtype. Boxplots show median and interquartile range; points show individual samples. (B) Variance distribution across retained methylation probes. Dashed line denotes the minimum variance among retained probes in the processed matrix.

For miRNA expression, $\log_2(\text{RPM} + 1)$-transformed values were used directly. Missing values in the miRNA matrix represent true non-detections (RPM = 0) and were therefore assigned a value of zero rather than imputed, preserving the biological interpretation of absent expression. miRNA features missing in more than 50% of samples were removed (1,312 of 1,965 features), a more permissive missingness threshold than applied to RPPA (20%), reflecting the tissue-specific and context-dependent expression patterns of mature miRNAs. Following low-expression filtering (median $\log_2(\text{RPM} + 1) < 1.0$), 200 features were retained by MAD (threshold: 0.60). The clustered heatmap of the 50 most variable retained miRNAs (Figure 1.3) illustrates broad subtype-associated expression patterns across the labelled cohort.

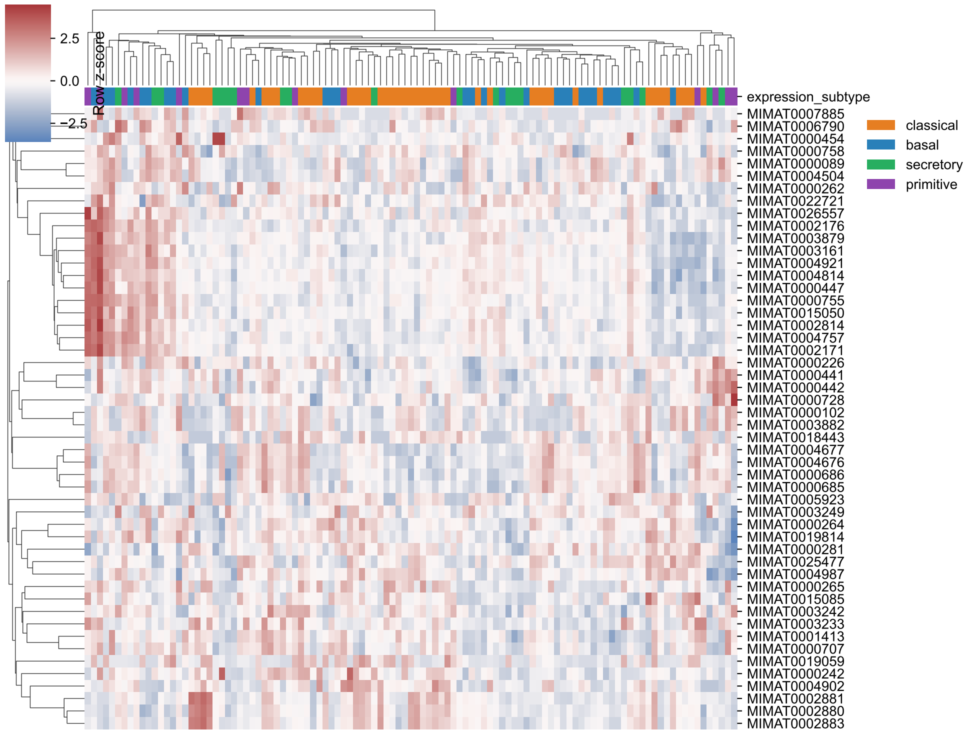


Figure 1.3: Clustered heatmap of 50 most variable miRNA features across labelled TCGA-Lung Squamous Cell Carcinoma (LUSC) samples (n = 107). Rows are miRNA features and columns are samples; values are row-scaled z-scores.

For RPPA, the 238 antibody-level protein abundance estimates produced by the University of Texas MD Anderson Cancer Center (MDACC) TCGA proteome characterisation centre were used directly following removal of 38 antibodies with > 20% missing values across the labelled cohort; no MAD filtering was applied, as the feature count was already within a manageable range for the available sample size. The antibody-level correlation structure (Figure 1.4) revealed distinct co-regulation clusters annotated by the subtype in which each antibody showed the highest mean abundance, consistent with the known proteomic differences between LUSC molecular subtypes.

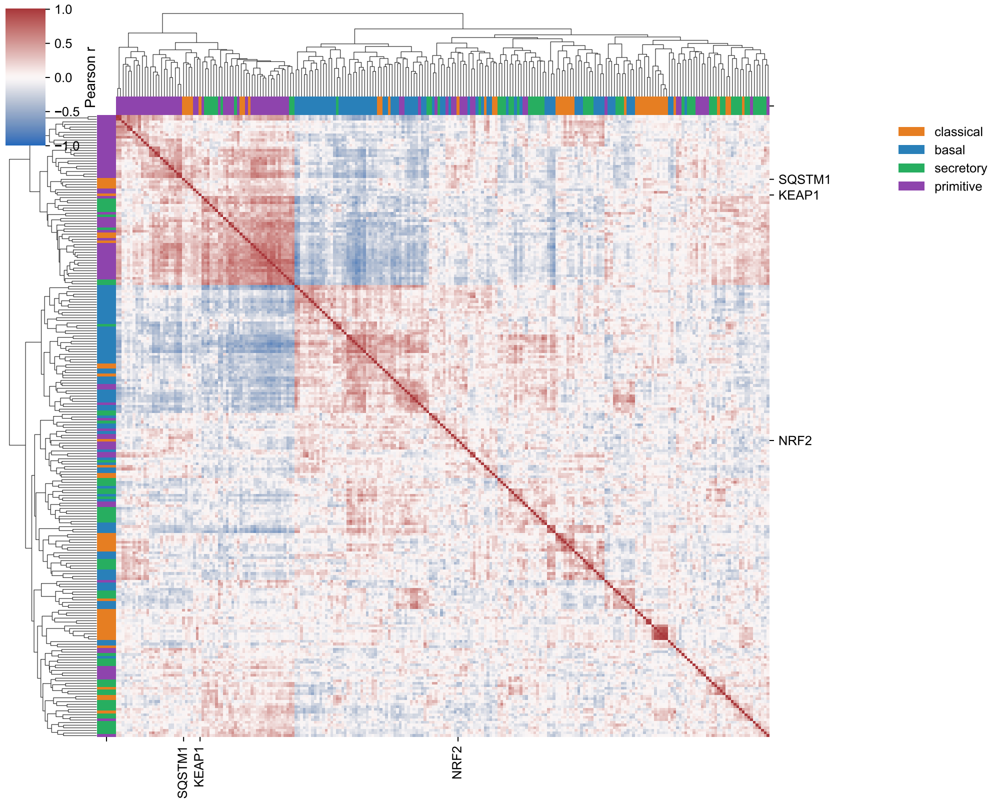


Figure 1.4: Hierarchically-clustered correlation heatmap across all 238 RPPA antibodies (n = 112 labelled samples). Rows and columns represent antibodies; colour denotes pairwise antibody-antibody correlation across RPPA samples. Annotation strips indicate the expression subtype in which each antibody had the highest mean abundance.

For CNV, GISTIC2-thresholded discrete values on the integer scale $\{-2, -1, 0, 1, 2\}$ were used without further transformation, as normalisation is not applicable to ordinal copy number calls. The top 2,000 most variable genes were selected by MAD (threshold: 1.00); a threshold of 1.00 corresponds to the first step in the discrete MAD distribution for GISTIC2 data, providing a natural and data-driven justification for this cutoff. Across the retained 2,000 features, the fraction of the processed CNV matrix with non-zero GISTIC2-thresholded states (i.e., any degree of copy number alteration) varied across subtypes (Figure 1.5), reflecting differences in relative genomic alteration burden within this feature set.

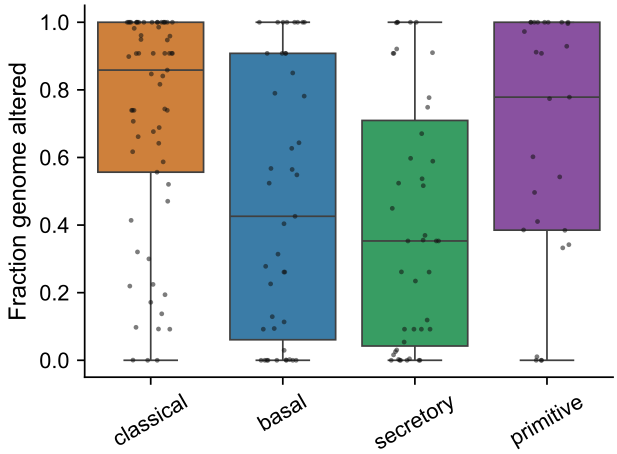


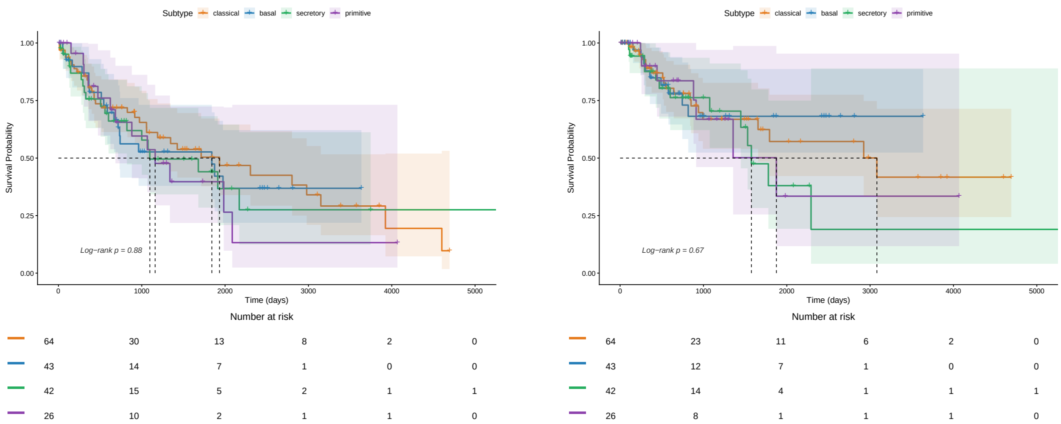
Figure 1.5: Fraction of retained CNV features with non-zero GISTIC2-thresholded copy-number state per sample, stratified by subtype (n = 178). Boxplots show median and interquartile range; points show individual samples. As the calculation is based on the 2,000 retained CNV features, values represent relative alteration burden within the processed CNV matrix rather than genome-wide Fraction of the Genome Altered (FGA).

1.3 In silico Phenotypic Characterisation of LUSC Subtypes

To establish the biological validity and clinical relevance of downstream classification, the LUSC molecular subtypes were characterised across three independent axes: clinical outcome (Section 1.3.1), tumour immune microenvironment composition (Section 1.3.2), and somatic mutation landscape (Section 1.3.3). These analyses are performed prior to any supervised modelling and are independent of the omics feature spaces used for classification.

1.3.1 Clinical Characteristics

Kaplan-Meier survival curves for Overall Survival (OS) and Progression-Free Interval (PFI) across the four LUSC subtypes are shown in Figure 1.6. PFI was selected as the primary endpoint consistent with TCGA clinical data resource recommendations for LUSC, in which PFI is considered a more reliable endpoint than OS.



(a) Overall Survival

(b) Progression-Free Interval

Figure 1.6: Survival curves showing Overall Survival (a) and Progression-Free Interval (b) for TCGA-LUSC samples (n=178) stratified by molecular subtypes. Median survival and 95% confidence intervals were estimated by the Brookmeyer-Crowley method. Shaded regions denote 95% confidence intervals and risk tables show the number of patients remaining at risk at selected time points.

Secretory and primitive subtype show the shortest estimated median OS of 36.0 and 38.1 months respectively. However, global log-rank tests did not reach statistical significance for either OS (log-rank $p = 0.88$) or PFI (log-rank $p = 0.67$), indicating that survival differences are not reliably distinguishable in the current cohort. The wide, overlapping 95% confidence intervals visible in Figure 1.6 are consistent with limited power due to small per-subtype sample sizes, particularly for the primitive subtype ($n = 26$). These results reflect the reduced statistical power of the labelled subcohort used in this study relative to the full TCGA-LUSC dataset, and do not meaningfully contradict prior reports of survival differences between LUSC molecular subtypes (Hammerman et al., 2012; Wilkerson et al., 2010).

Cox proportional hazards regression, using the classical subtype as the reference category, is shown in Figure 1.7. Hazard ratio estimates and their 95% confidence intervals are reported; the proportional hazards assumption was assessed using scaled Schoenfeld residuals.

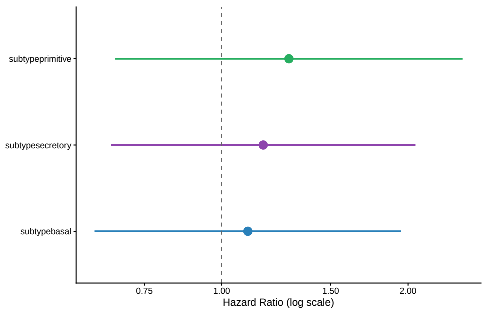


Figure 1.7: Hazard ratios for overall survival by LUSC molecular subtype from Cox proportional hazards regression, with classical subtype as reference (n=AVOCADO). Points denote hazard ratios and horizontal bars represent 95% confidence intervals. HR > 1 indicates worse prognosis relative to classical subtype. The proportional hazards assumption was verified using scaled Schoenfeld residuals (cox.zph global p=AVOCADO). Concordance index=AVOCADO

1.3.2 Immune Microenvironment

Tumour immune microenvironment composition was estimated using two complementary deconvolution approaches applied to gene expression data: EPIC (Racle et al., 2017) for cell-fraction estimation and xCell (Aran et al., 2017) for enrichment scoring, following the method selection guidance of Sturm *et al.* (Sturm et al., 2019).

Mean EPIC cell fractions stratified by molecular subtype are shown in Figure 1.8. All cell compartments showed statistically significant variation across subtypes by Kruskal-Wallis test after Benjamini-Hochberg correction ($q < 0.001$), with the exception of B cells (q = ns). The *otherCells* compartment, representing tumour cells and uncharacterised non-immune populations, was the dominant fraction across all subtypes (mean: 0.37-0.63), as expected for bulk tumour samples. The primitive subtype showed the highest *otherCells* fraction (mean = 0.625), consistent with the high tumour cellularity and reduced immune infiltrate associated with the poorly differentiated, proliferative biology of this subtype (Hammerman et al., 2012; Wilkerson et al., 2010). In contrast, the secretory subtype showed the lowest *otherCells* fraction (mean = 0.368) and the highest Cancer-Associated Fibroblast (CAF) fraction (mean = 0.431), indicating a substantially more reactive stromal microenvironment. Classical tumours showed the highest Cluster of Differentiation 4 (CD4) T cell fraction (mean = 0.104) and the second-highest *otherCells* fraction (mean = 0.620), suggesting a predominantly tumour-cell-dense composition with a moderate adaptive immune component.

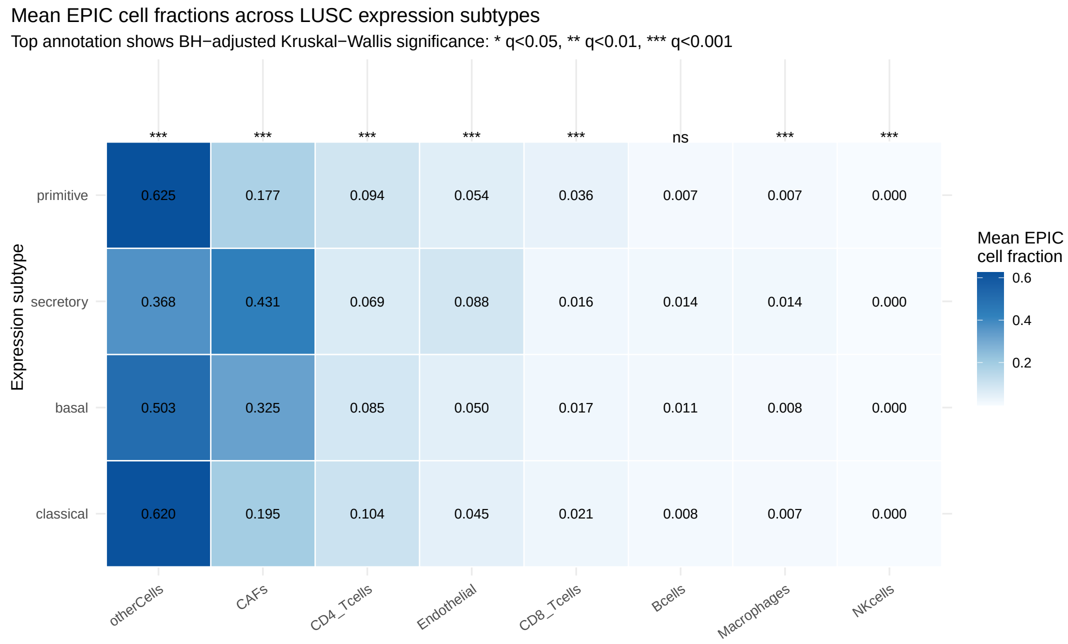


Figure 1.8: Mean EPIC cell fractions across LUSC subtypes ($n = 178$). Rows represent expression subtypes; columns represent EPIC-inferred cell compartments ordered by mean fraction. Numeric values show the mean fraction per subtype-cell-type combination. Significance annotations denote BH-adjusted Kruskal-Wallis q-values across subtypes ($*q < 0.05$, $**q < 0.01$, $***q < 0.001$; ns: not significant).

Secretory subtype showed the highest EPIC fractions for macrophages and CAFs; primitive subtype showed the highest fraction of uncharacterised cells (otherCells), consistent with its highly proliferative phenotype. B cells did not differ significantly across subtypes.

Cluster of Differentiation 8 (CD8) T cell fractions (EPIC) were highest in the primitive subtype (mean = 0.036) and lowest in the secretory subtype (mean = 0.016). Macrophage fractions were highest in the secretory subtype (mean = 0.014, $q < 0.001$) and significantly elevated relative to all other subtypes after Benjamini-Hochberg correction. Selected cell-type distributions are shown in Figure 1.9.

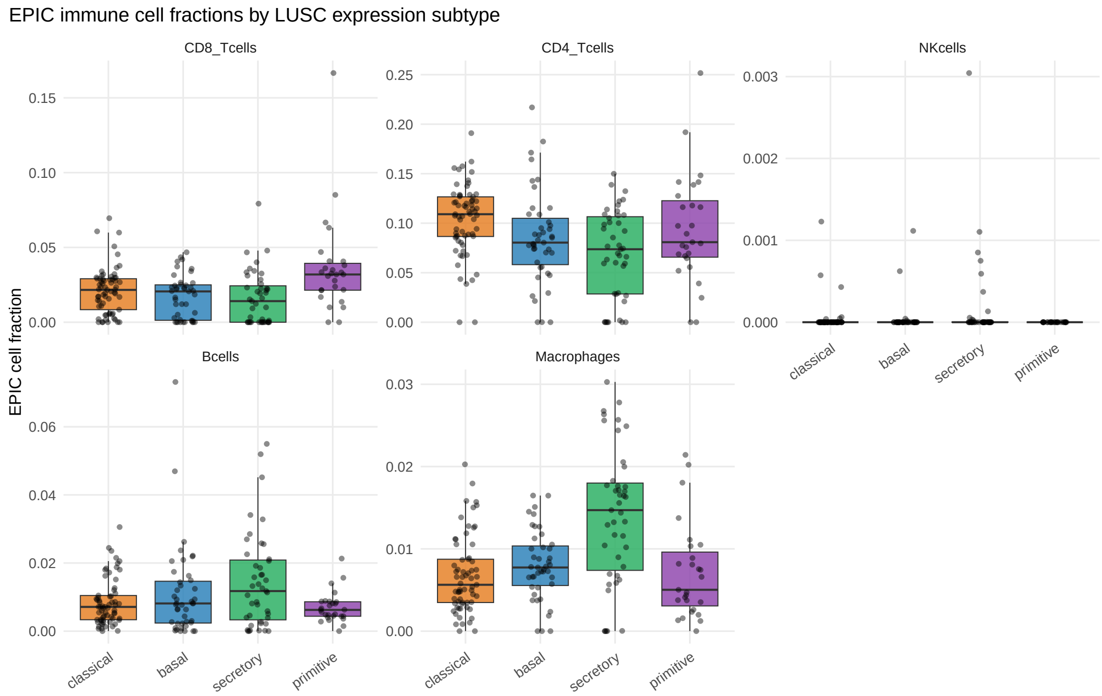


Figure 1.9: EPIC-estimated cell fractions for five cell populations (CD8 T cells, CD4 T cells, Natural Killer (NK) cells, B cells, macrophages) across subtypes. Boxplots show median and interquartile range; individual points are samples.

xCell enrichment scores for selected immune populations are shown in Figure 1.10. Secretory subtype showed the highest enrichment for macrophages M1 and M2, B cells, class-switched memory B cells, CD8 T cells, CD8 central memory T cells, and Tregs (all $q < 0.05$ after Benjamini-Hochberg correction). Broad NK cell enrichment scores did not differ significantly across subtypes by xCell ($q = 0.642$). EPIC assigned highest CD8 fraction to the primitive subtype while xCell assigned highest enrichment to the secretory subtype, reflecting known methodological differences between cell-fraction deconvolution (EPIC) and gene set enrichment scoring (xCell).

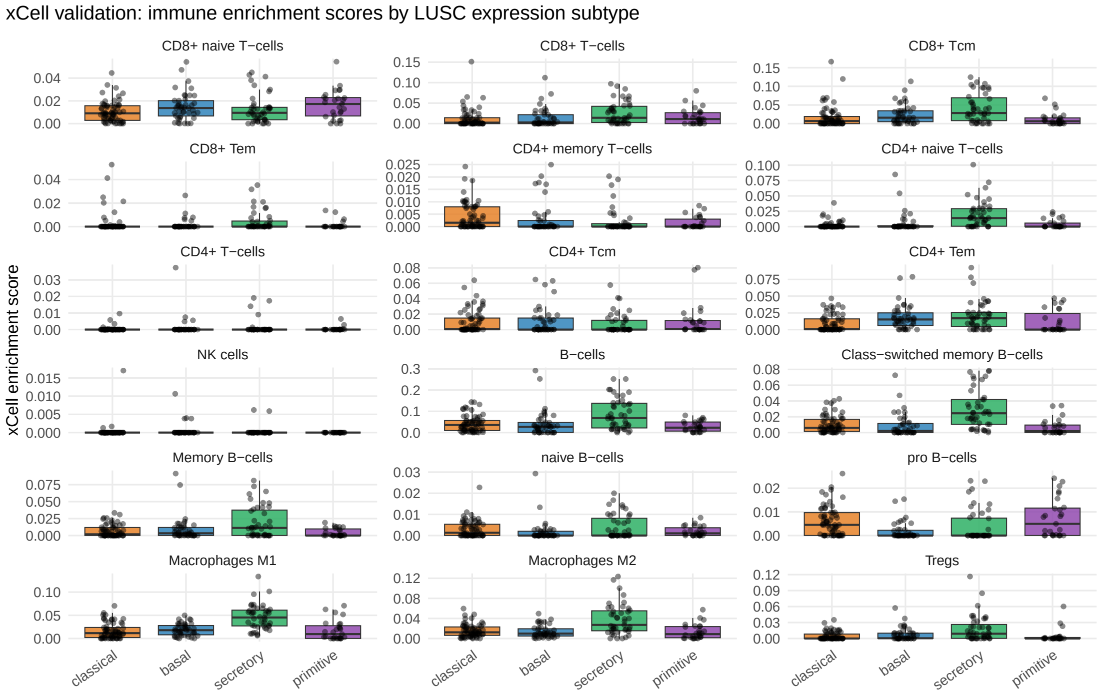


Figure 1.10: xCell enrichment scores for selected immune populations across LUSC molecular subtypes (n = 178). xCell scores are relative enrichment estimates and not directly comparable across cell types.

1.3.3 Somatic Mutation Landscape

Somatic mutation landscape across a curated panel of LUSC driver genes is shown in Figure 1.11. Of 174 samples with available mutation data, 157 (90.2%) harboured at least one alteration in the curated driver panel, reflecting the high overall mutation burden characteristic of tobacco-associated squamous cell carcinoma.

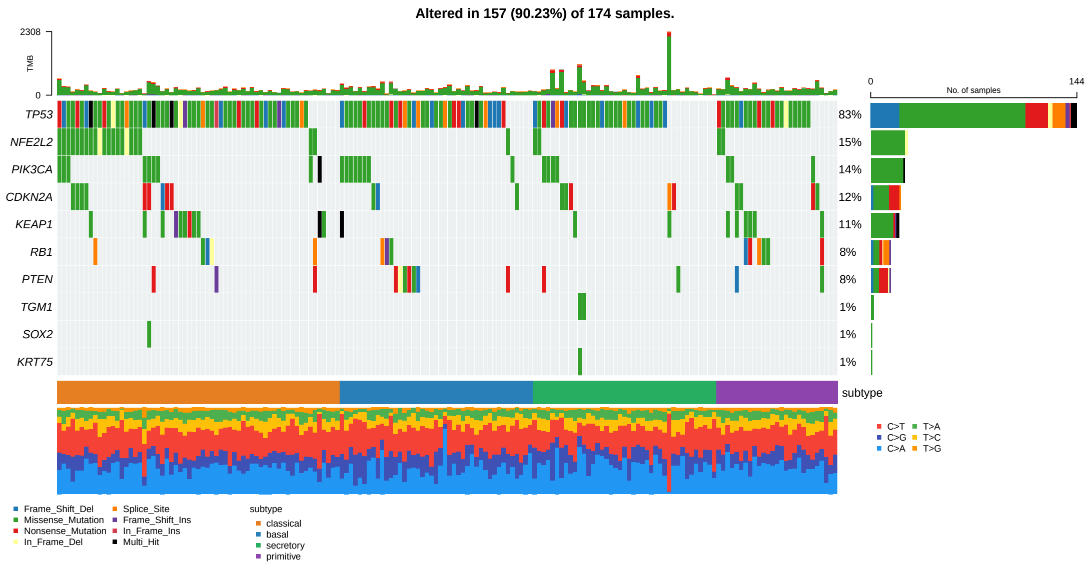


Figure 1.11: Oncoplot of somatic alterations across a curated LUSC driver gene panel in TCGA-LUSC samples with available mutation data (n = 174; altered in 157, 90.2%). Alteration frequencies (right margin) are computed across all samples.

TP53 was the most frequently mutated gene overall (83%) (Figure 1.11) but did not show subtype-specific enrichment, consistent with its role as a common subtype-independent driver alteration in LUSC and other tobacco-related squamous carcinomas. NFE2L2 (15%) and KEAP1 (11%) were notable among the remaining frequently altered genes; as components of the canonical oxidative stress response pathway, their co-occurrence reflects the high prevalence of this pathway in LUSC. PIK3CA (14%) and CDKN2A (12%) alterations were distributed across subtypes without significant enrichment in the current cohort. RB1 (8%) and PTEN (8%) were present at low frequencies; both are known to be altered in the primitive and basal subtypes in larger LUSC cohorts (Hammerman et al., 2012), though the statistical power to detect subtype-specific enrichment at these frequencies is limited by the current cohort size. The squamous lineage markers TGM1, SOX2, and KRT75 were each somatically altered in approximately 1% of samples.

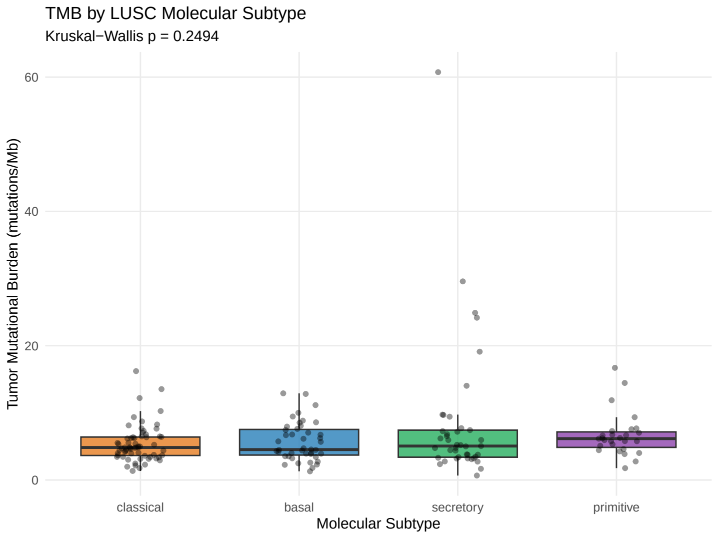


Figure 1.12: Tumour mutational burden (Tumor Mutational Burden (TMB)) per LUSC molecular subtype (n = 174). Boxplots show median and interquartile range; individual points are samples. TMB did not differ significantly across subtypes by Kruskal-Wallis test.

TMB did not vary significantly across subtypes (Figure 1.12), indicating that the molecular distinctions between LUSC subtypes are not attributable to differential overall somatic mutation burden. This finding is consistent with the ubiquitous tobacco mutational signature across LUSC subtypes, which imposes a high and relatively uniform TMB background (Alexandrov et al., 2013).

1.4 Dimensionality Structure and Subtype Separability

Prior to supervised classification, geometric separability of the four LUSC molecular subtypes was assessed for each omics feature space using PCA, Silhouette scoring, and PERMANOVA.

1.4.1 Dimensionality Structure

The number of principal components required to explain 80% of cumulative variance varied substantially across platforms, reflecting differences in feature covariance structure across omics modalities. Gene expression (n=178, 2000 genes) required the highest number of (62 PCs; PC1 = 12.3% variance), consistent with transcriptional variation being distributed diffusely across thousands of co-regulated gene programs. miRNA (n=107, 200 features) required 30 components (PC1 = 16.6% variance) and RPPA (n=112, 238 proteins) required 27 components (PC1 = 16.4% variance). In contrast, DNA methylation (n=76, 2000 probes) required only 11 components to reach 80% variance, with

PC1 explaining an unusually high 57.7% of total variance. This likely represents genome-wide methylation differences between samples, analogous to CpG Island Methylator Phenotype (CIMP)-like methylation architecture reported in squamous carcinomas. CNV (n=178, 2000 genes) was the most compact of all modalities, requiring only 5 components (PC1 = 44.7%; PC2 = 20.8%), reflecting coarse-chromosomal-arm-scale nature of copy number events relative to gene-level expression or methylation variation.

Table 1.3: Dimensionality structure and subtype separability for each omics modality. PCs denotes the number of principal components required to explain ≥ 80% of cumulative variance. Silhouette score (S) and PERMANOVA statistics were computed on this reduced represeantation. Per-subtype silhouette scores are shown in parentheses (Cl: classical; Ba: basal; Se: secretory; Pr: primitive). $^{**}p \leq 0.001$; $^{*}p = 0.005$. The minimum achievable permutation p-value with 999 permutations is 0.001; values reported as ≤ 0.001 indicate zero permutations produced a pseudo-F as extreme as the observed statistic.

Modality	**n**	**Feat.**	**PCs**	**PC1 (%)**	**S**	**Per-subtype S**				**pseudo-F**	**p**
						Cl	**Ba**	**Se**	**Pr**		
Gene expression	178	2000	62	12.3	0.047	0.091	0.070	0.025	−0.062	8.997	$\leq 0.001^{**}$
miRNA expression	107	200	30	16.6	0.031	0.098	0.028	0.006	−0.133	5.191	$\leq 0.001^{**}$
RPPA	112	238	27	16.4	0.008	−0.008	0.055	0.018	−0.047	3.935	$\leq 0.001^{**}$
DNA methylation	76	2000	11	57.7	−0.060	−0.128	−0.028	−0.016	−0.011	3.491	0.005^{*}
CNV	178	2000	5	44.7	−0.031	0.087	−0.106	−0.011	−0.224	6.759	$\leq 0.001^{**}$

1.4.2 Subtype Separability

PERMANOVA. All modalities produced statistically significant subtype group separation by PERMANOVA (p ≤ 0.005 in all cases; Table 1.3), indicating that subtype group centroids are distinguishable from random label permutations across every omics platform. Gene expression showed the strongest centroid separation (pseudo-F = 8.997, $p \leq 0.001$), followed by CNV (pseudo-F = 6.759, $p \leq 0.001$), miRNA (pseudo-F = 5.191, $p \leq 0.001$), RPPA (pseudo-F = 3.935, $p \leq 0.001$), and methylation (pseudo-F = 3.491, $p = 0.005$).

Silhouette scores. Mean silhouette scores revealed different modality ordering from PERMANOVA, with gene expression achieving the highest individual-sample cluster cohesion ($S = 0.047$), followed by miRNA ($S = 0.031$), RPPA ($S = 0.008$), CNV ($S = -0.031$), and methylation ($S = -0.060$). Negative silhouette scores for CNV and methylation indicate that, on average, samples are geometrically closer to an adjacent subtype's centroid than their own, reflecting high within-group variance relative to between-group separation, even when centroids are statistically distinguishable.

Per-subtype Silhouette scores. Primitive subtype showed negative mean silhouette scores across all modalities (GE: −0.062; miRNA: −0.133; RPPA: −0.047; methylation: −0.011; CNV: −0.224), indicating that primitive samples are consistently closer, on average, to the centroid of another subtype than to their own. Classical and basal subtypes showed highest silhouette scores in majority of modalities, suggesting more internally consistent omics feature profiles.

CNV structural paradox. The second-highest PERMANOVA pseudo-F for CNV (6.759) is accompanied by a negative mean silhouette (−0.031) and the lowest per-class F1 in classification (Section 1.5). The high compactness of the CNV PCA structure (5 components for 80% variance; PC1 = 44.7%) suggests that the majority of CNV variation is dominated by a small number of large chromosomal events- likely including focal amplifications at 3q26 (encompassing *SOX2* and *PIK3CA*) and 3p deletions characteristic of LUSC. The negative per-subtype silhouette for basal (−0.106), secretory (−0.011), and primitive (−0.224) subtypes within the CNV feature space confirms that copy number profiles do not encode the four-subtype structure at an individual-sample level, consistent with molecular subtypes being primarily defined by transcriptional and epigenetic programmes rather than structural genomic events.

Per-modality PCA scatter plots for each of the five omics modalities are shown in Figures 1.13-1.17. These projections capture 12.3-65.3% of total per-modality variance.

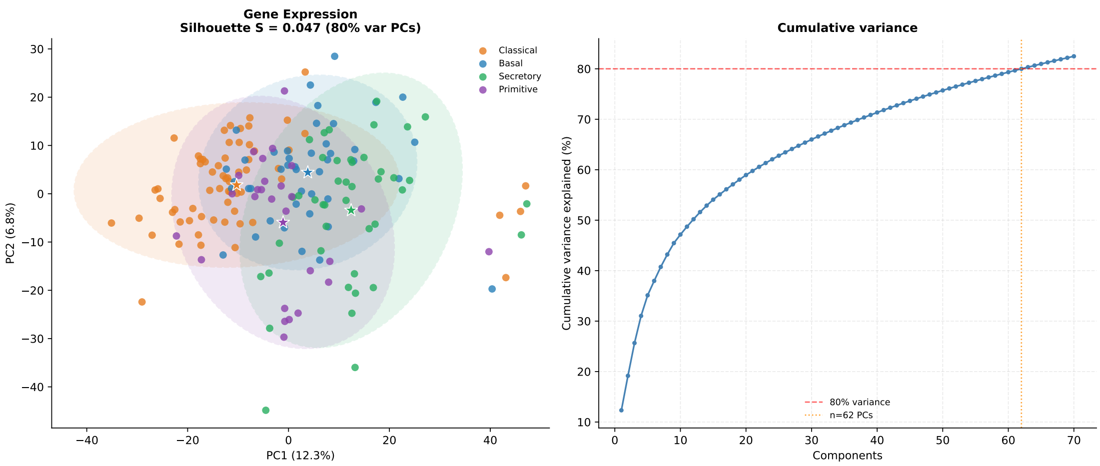


Figure 1.13: PCA of gene expression feature space (n = 178; 2,000 MAD-selected genes); Left: PC1 versus PC2. Principal Component (PC) variance contributions shown on axis labels. Right: cumulative variance explained by top principal components.

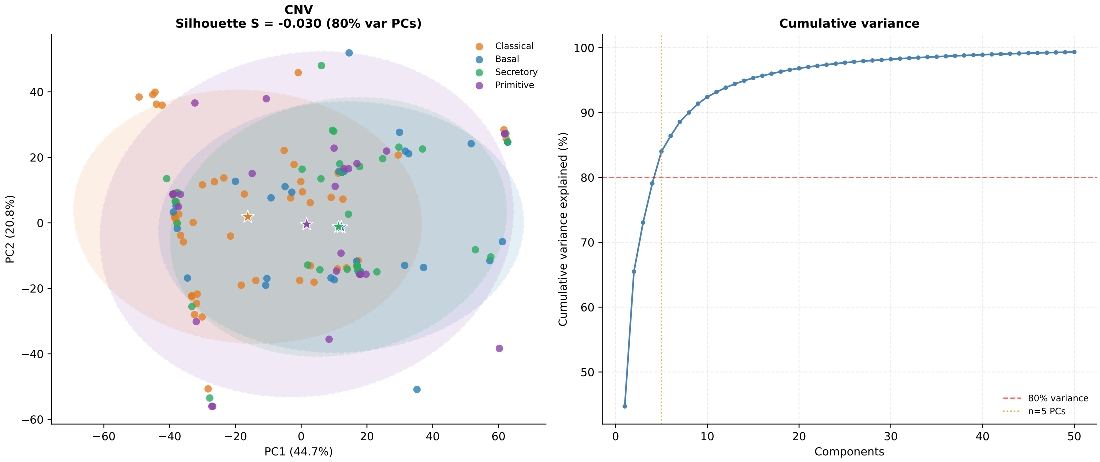


Figure 1.14: PCA of CNV feature space ($n = 178$; 2,000 MAD-selected genes); Left: PC1 versus PC2. PC variance contributions shown on axis labels. Right: cumulative variance explained by top principal components.

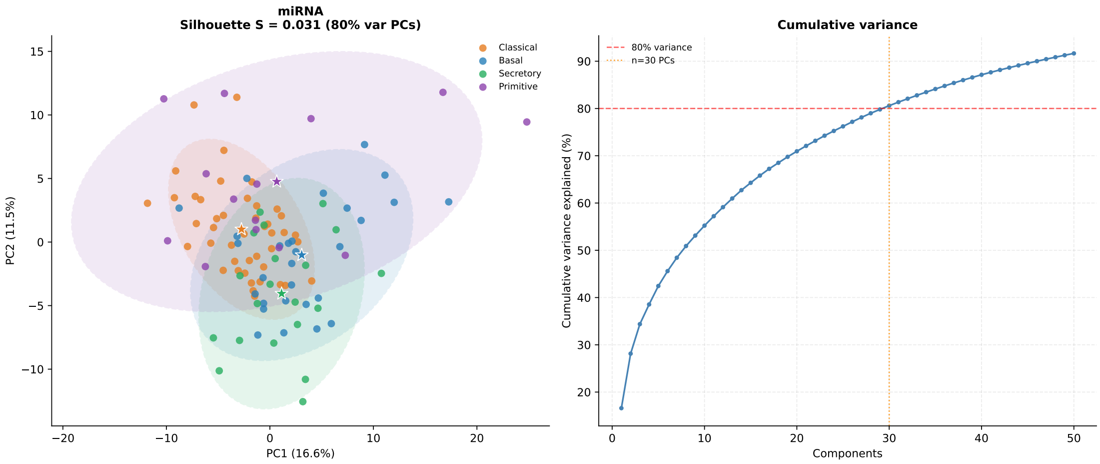


Figure 1.15: PCA of miRNA feature space ($n = 107$; 200 MAD-selected mature miRNAs; $\log_2(\text{RPM} + 1)$); Left: PC1 versus PC2. PC variance contributions shown on axis labels. Right: cumulative variance explained by top principal components.

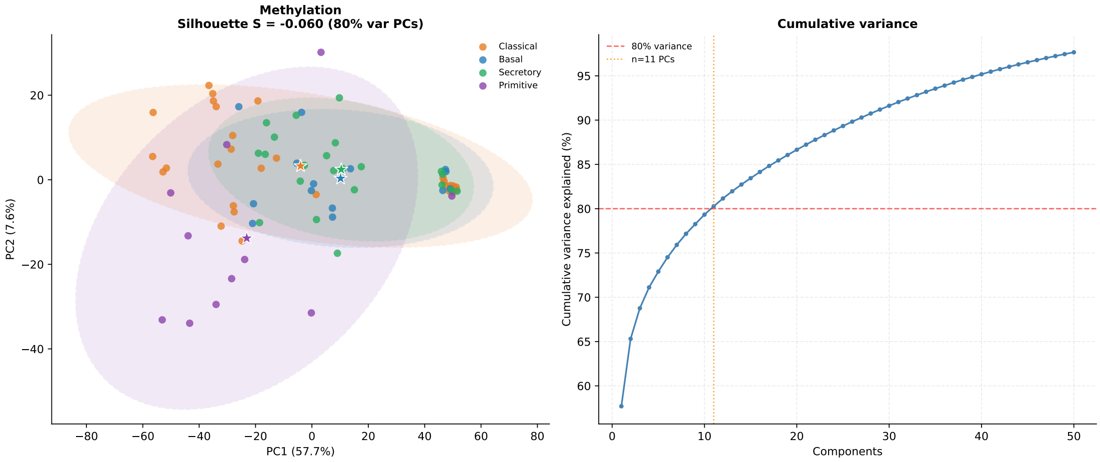


Figure 1.16: PCA of DNA methylation feature space (n = 76; 2,000 MAD-selected HumanMethylation450 probes; beta values). Left: PC1 versus PC2. PC variance contributions shown on axis labels. Right: cumulative variance explained by top principal components.

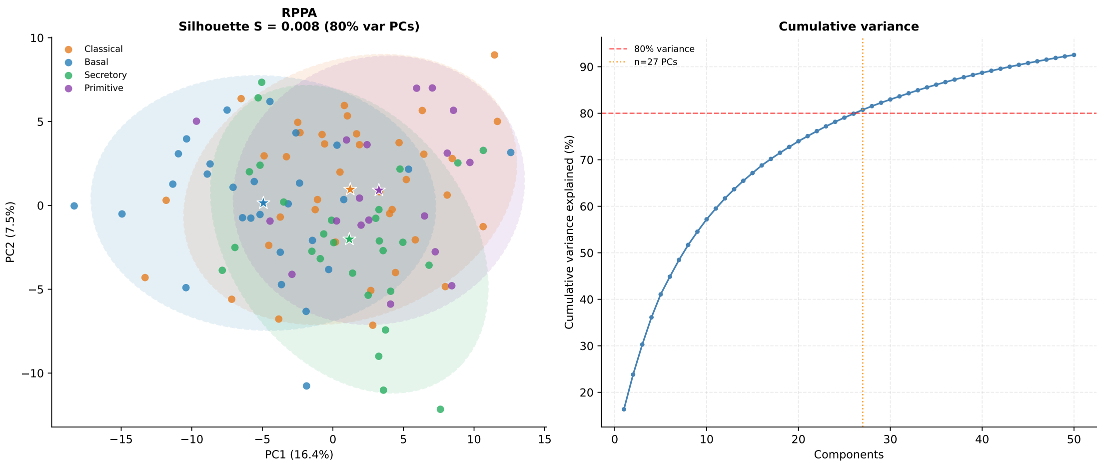


Figure 1.17: PCA of RPPA feature space (n = 112; 238 antibodies). Left: PC1 versus PC2. PC variance contributions shown on axis labels. Right: cumulative variance explained by top principal components.

1.5 Performance Across Omics Modalities

1.6 SHAP Feature Importance and Stability

1.7 Cross-Model Feature Overlap

1.8 Cross-Modality Feature Overlap

1.9 Pathway Enrichment

References

Alexandrov, L. B., Nik-Zainal, S., Wedge, D. C., Aparicio, S. A. J. R., Behjati, S., Biankin, A. V., Bignell, G. R., Bolli, N., Borg, A., Børresen-Dale, A.-L., Boyault, S., Burkhardt, B., Butler, A. P., Caldas, C., Davies, H. R., Desmedt, C., Eils, R., Eyfjörd, J. E., Foekens, J. A., … Stratton, M. R. (2013). Signatures of mutational processes in human cancer. *Nature*, *500*(7463), 415–421. https://doi.org/https://doi.org/10.1038/nature12477

Aran, D., Hu, Z., & Butte, A. J. (2017). Xcell: Digitally portraying the tissue cellular heterogeneity landscape. *Genome Biol.*, *18*(1), 220. https://doi.org/https://doi.org/10.1186/s13059-017-1349-1

Hammerman, P. S., Lawrence, M. S., Voet, D., Jing, R., Cibulskis, K., Sivachenko, A., Stojanov, P., McKenna, A., Lander, E. S., Gabriel, S., Getz, G., Sougnez, C., Imielinski, M., Helman, E., Hernandez, B., Pho, N. H., Meyerson, M., Chu, A., Chun, H.-J. E., … group, D. w. (2012). Comprehensive genomic characterization of squamous cell lung cancers. *Nature*, *489*(7417), 519–525. https://doi.org/10.1038/nature11404

Racle, J., de Jonge, K., Baumgaertner, P., Speiser, D. E., & Gfeller, D. (2017). Simultaneous enumeration of cancer and immune cell types from bulk tumor gene expression data (A. Valencia, Ed.). *eLife*, *6*, e26476. https://doi.org/10.7554/eLife.26476

Sturm, G., Finotello, F., Petitprez, F., Zhang, J. D., Baumbach, J., Fridman, W. H., List, M., & Aneichyk, T. (2019). Comprehensive evaluation of transcriptome-based cell-type quantification methods for immuno-oncology. *Bioinformatics*, *35*(14), i436–i445. https://doi.org/10.1093/bioinformatics/btz363

Wilkerson, M. D., Yin, X., Hoadley, K. A., Liu, Y., Hayward, M. C., Cabanski, C. R., Muldrew, K., Miller, C. R., Randell, S. H., Socinski, M. A., Parsons, A. M., Funkhouser, W. K., Lee, C. B., Roberts, P. J., Dang, T. P., Parejko, M. K., & Hayes, D. N. (2010). Lung squamous cell carcinoma mrna expression subtypes are reproducible, clinically important, and correspond to normal cell types. *Clinical Cancer Research*, *16*(19), 4864–4875. https://doi.org/10.1158/1078-0432.CCR-10-0199